

Technical Document #790: Data Engineering - Comprehensive Overview

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Data lakes store raw data in its native format. They support structured, semi-structured, and unstructured data at any scale.

Schema evolution handles changes in data structure over time. Schema registries manage schema versions and ensure backward compatibility.

ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loads it into a target data store.

Partitioning divides data into smaller, more manageable pieces. It improves query performance by reducing the amount of data scanned.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Key Takeaways:

- ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loads it into a target data store.
- Data quality ensures data is accurate, complete, and consistent.
- Data pipelines automate the movement and transformation of data between systems.